

SOC-Based Phishing Email Threat Investigation

Project Overview

Conducted a SOC-style phishing email investigation involving email header analysis, malicious URL analysis, IOC identification, and threat intelligence correlation. The investigation focused on detecting credential harvesting attempts and documenting mitigation strategies using industry-standard cybersecurity tools.

Resume Highlights

- Conducted phishing email investigations using VirusTotal, MXToolbox, and URLScan.io.
- Performed email header analysis including SPF, DKIM, and DMARC validation checks.
- Identified Indicators of Compromise (IOCs) and investigated malicious infrastructure.
- Mapped phishing techniques to MITRE ATT&CK; framework.
- Documented mitigation strategies and incident response recommendations.
- Analyzed spoofed domains and credential harvesting attempts.

Tools & Technologies Used

Tool	Purpose
VirusTotal	Threat intelligence & URL analysis
MXToolbox	Email header analysis
URLScan.io	Malicious URL investigation
Wireshark	Packet analysis
Any.Run	Sandbox execution
Kali Linux	Security testing environment

Threat Scenario

Investigated a phishing email impersonating Microsoft Security Team requesting an urgent password reset. The email contained spoofed branding and malicious credential harvesting links designed to steal user credentials.

Technical Findings

- SPF, DKIM, and DMARC authentication failures detected.
- Malicious phishing domain identified and verified using threat intelligence tools.
- Credential harvesting behavior confirmed through sandbox analysis.
- Suspicious infrastructure and spoofed sender domains detected.

Indicators of Compromise (IOCs)

IOC Type	Indicator
Malicious Domain	microsoft-login-security.ru

Attack Type	Credential Harvesting
Threat Category	Phishing
Risk Severity	High

MITRE ATT&CK; Mapping

Technique	MITRE ID
Phishing	T1566
Credential Harvesting	T1056
Spearphishing Link	T1566.002

Skills Demonstrated

Threat Detection • Incident Response • IOC Analysis • Email Security • Threat Intelligence • SOC Operations • Phishing Analysis • Risk Assessment • Security Monitoring

Conclusion

This project demonstrates practical SOC Analyst skills through phishing email analysis, threat investigation, IOC identification, and security reporting. The investigation successfully identified a credential harvesting phishing attack and provided actionable remediation recommendations to improve organizational security posture.